Cost Analysis

2026-07-27

Contents

1 Assumptions

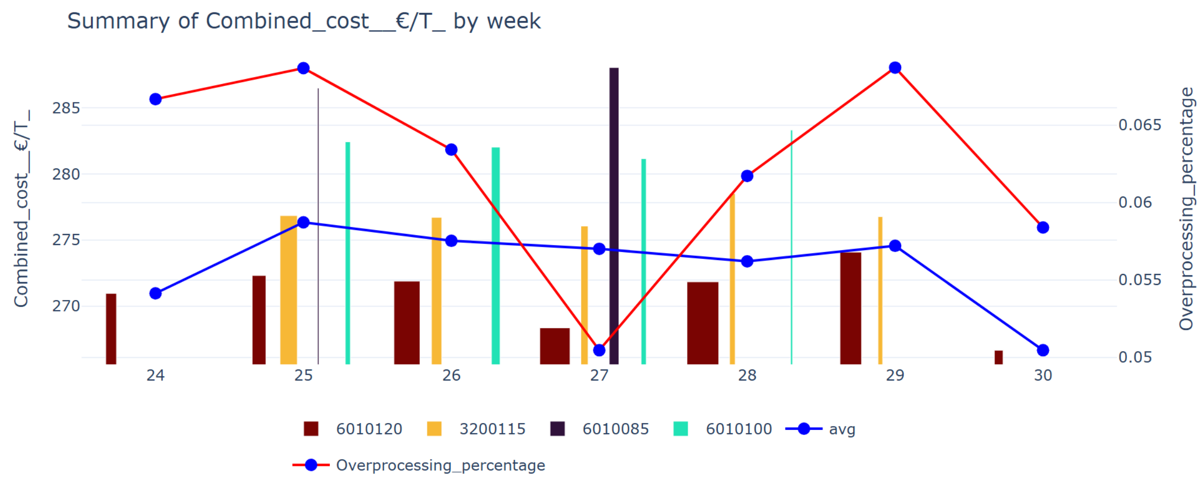
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

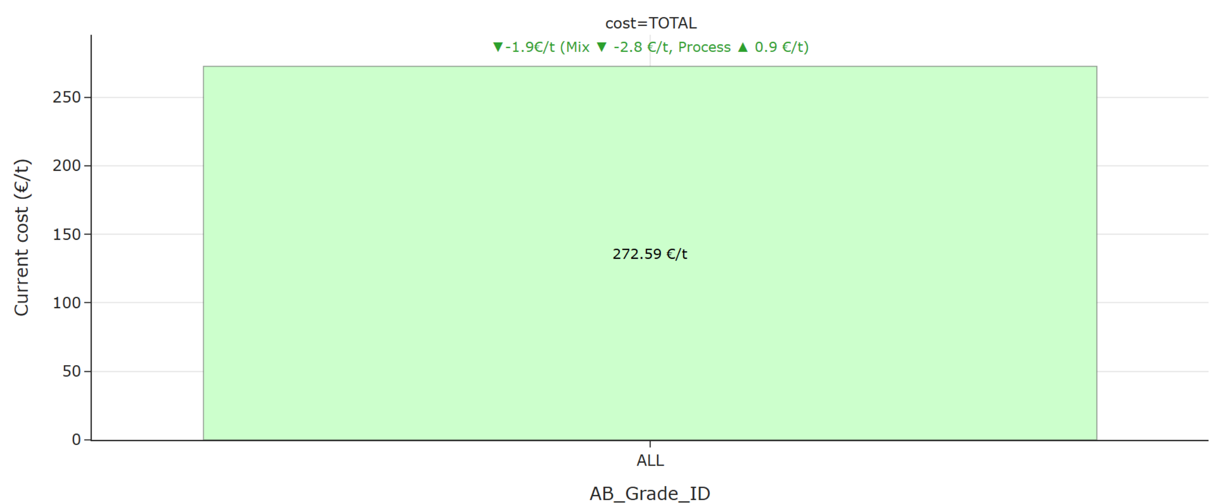
- **Period Analyzed**: June 14, 2026 – July 20, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.
  - **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
  - **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

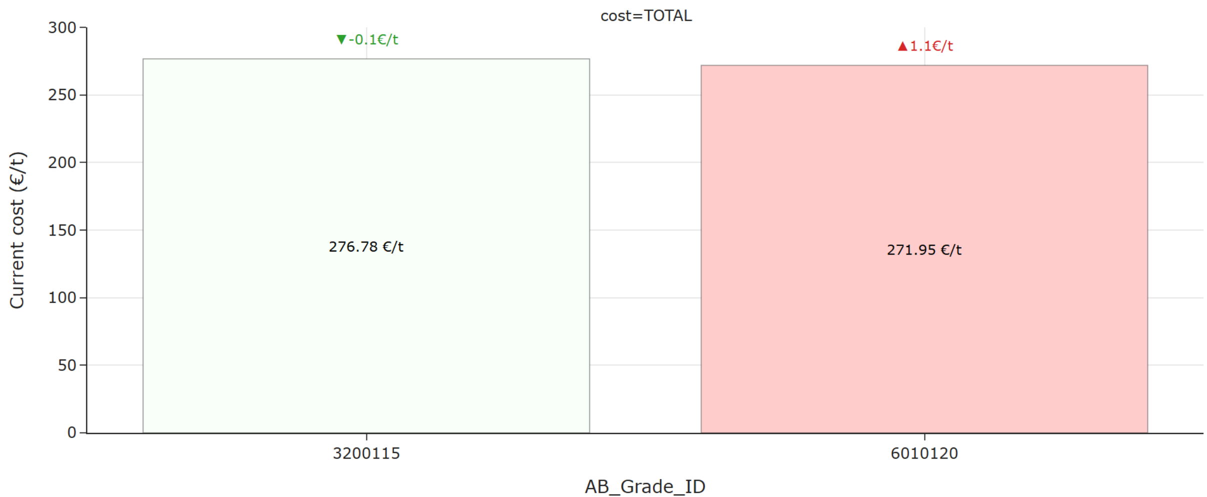
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw a decrease of 1.91 €/t, resulting in 272.59 €/t.

Breaking down the overall cost change, -2.81 €/t was driven by a cheaper mix of grades, and +0.90 €/t was due to a deterioration in the process.



For grade 3200115, TOTAL cost saw a decrease of 0.13 €/t, resulting in 276.78 €/t.

For grade 6010120, TOTAL cost saw an increase of 1.06 €/t, resulting in 271.95 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 6010120 (+1.06 €/t) and the strongest cost decrease was observed for grade 3200115 (-0.13 €/t). These movements stand out and may require further investigation.

Average change by component:

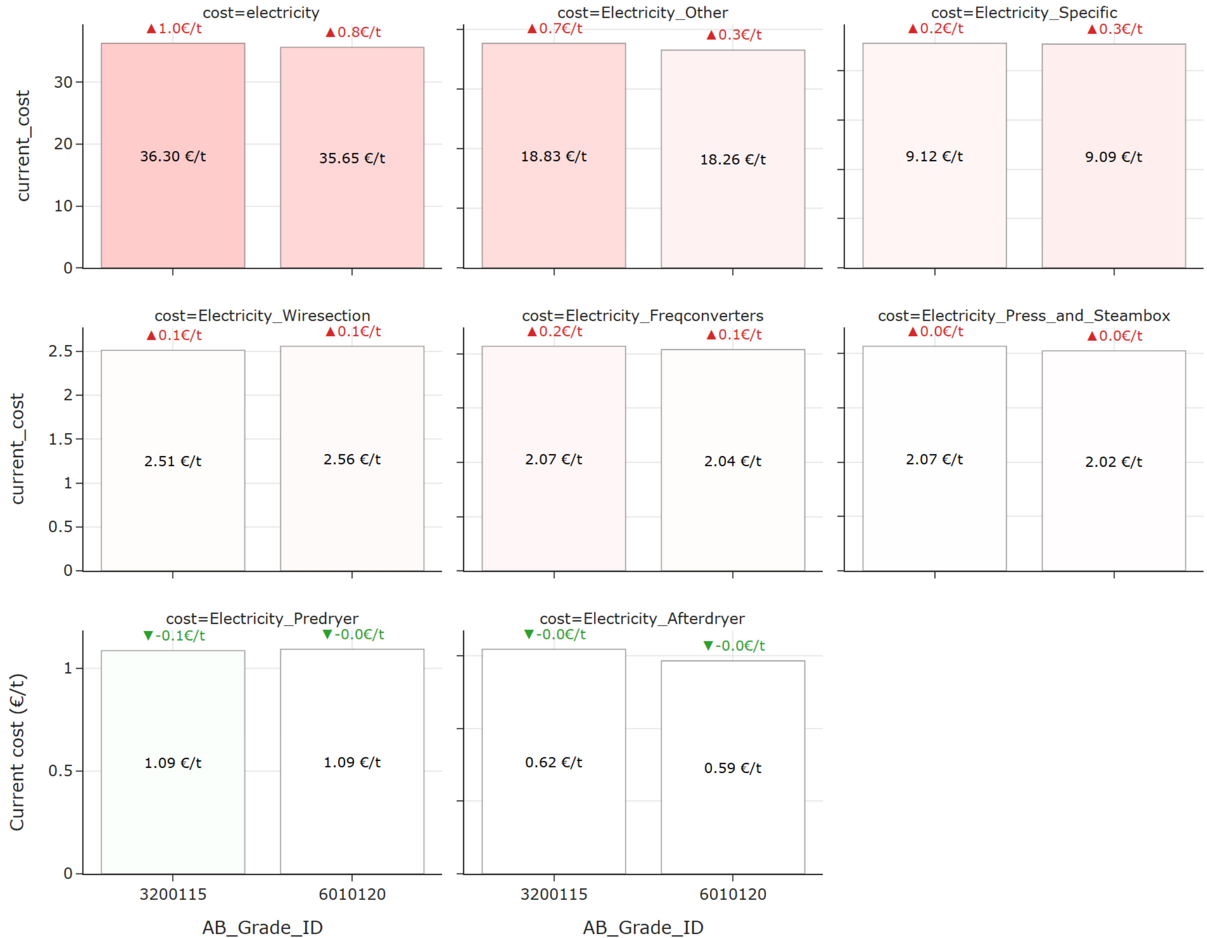
- **Steam__€/T** decreased on average (-1.55 €/t).
- **Electricity__€/T** increased on average (+0.90 €/t).
- **Starch__€/T** increased on average (+0.54 €/t).
- **Fibre_cost__€/T** increased on average (+0.43 €/t).
- **Chemicals__€/T** increased on average (+0.15 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity__€/T** (+0.90 €/t on average), with the highest increase in grade 3200115 (+1.00 €/t) and the largest average decrease comes from **Steam__€/T** (-1.55 €/t on average), with the biggest drop in grade 3200115 (-2.12 €/t). These components stand out and may require further investigation.

Average change by steam:

- **Steam__Predryers** decreased on average (-0.97 €/t).
- **Steam__Whitewater** decreased on average (-0.31 €/t).
- **Steam__Afterdryers** decreased on average (-0.25 €/t).
- **Steam__Heatexchangers** decreased on average (-0.04 €/t).
- **Steam__Starchkitchen** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam__Starchkitchen** (+0.01 €/t on average), with the highest increase in grade 3200115 (+0.03 €/t) and the largest average decrease comes from **Steam__Predryers** (-0.97 €/t on average), with the biggest drop in grade 3200115 (-1.91 €/t). These components stand out and may require further investigation.



Average change by electricity:

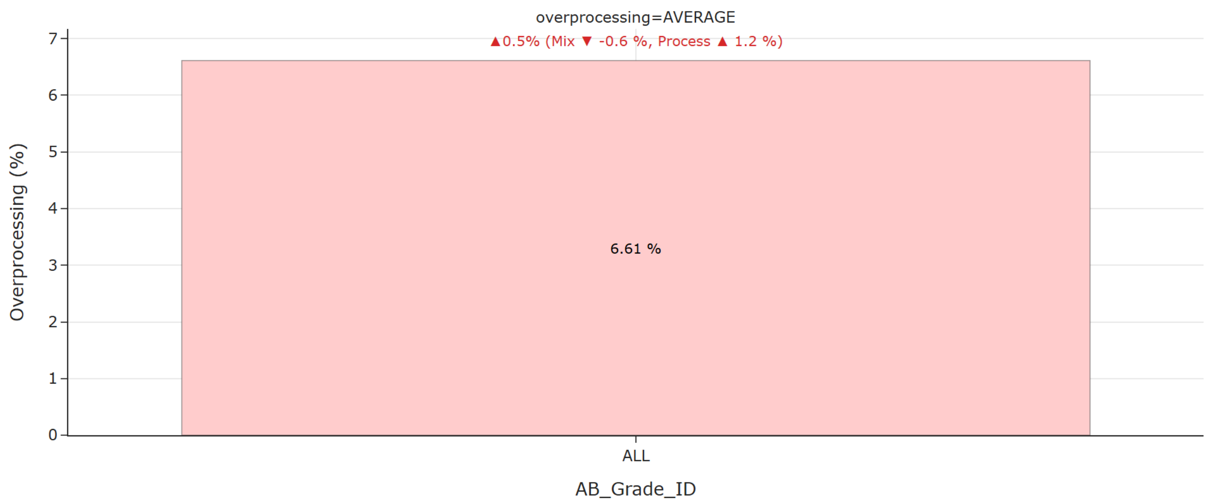
- **Electricity_Other** increased on average (+0.46 €/t).
- **Electricity_Specific** increased on average (+0.26 €/t).
- **Electricity_Freqconverters** increased on average (+0.11 €/t).
- **Electricity_Wiresection** increased on average (+0.08 €/t).
- **Electricity_Predryer** decreased on average (-0.04 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.03 €/t).
- **Electricity_Afterdryer** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.46 €/t on average), with the highest increase in grade 3200115 (+0.67 €/t) and the largest average decrease comes from **Electricity_Predryer** (-0.04 €/t on average), with the biggest drop in grade 3200115 (-0.07 €/t). These components stand out and may require further investigation.

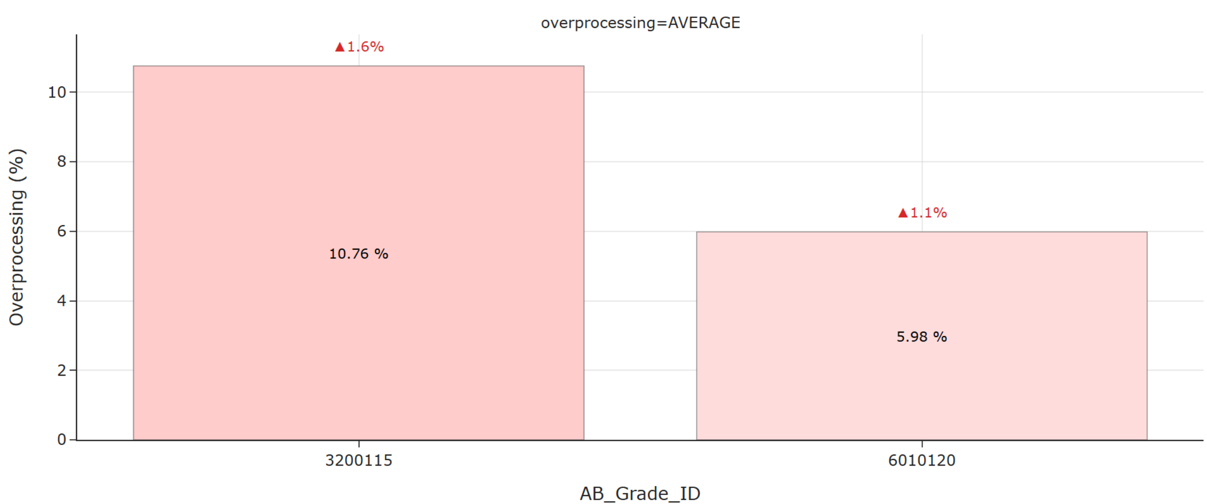
Average change by chemicals:

- **Sizing_Agent_*€/T*** increased on average (+0.10 €/t).
- **Fixative_2_*€/T*** increased on average (+0.09 €/t).
- **Bentonite_2_*€/T*** decreased on average (-0.09 €/t).
- **Retention_Aid_*€/T*** increased on average (+0.08 €/t).
- **Natriumhydroxide_*€/T*** decreased on average (-0.05 €/t).
- **Bentonite_1_*€/T*** increased on average (+0.03 €/t).
- **Defoamer_*€/T*** decreased on average (-0.02 €/t).
- **Deaerator_*€/T*** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.10 €/t on average), with the highest increase in grade 3200115 (+0.17 €/t) and the largest average decrease comes from **Bentonite_2_€/T** (-0.09 €/t on average), with the biggest drop in grade 3200115 (-0.09 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.55 %, resulting in 6.61 %.



For grade 3200115, TOTAL overprocessing saw an increase of 1.60 %, resulting in 10.76 %.

For grade 6010120, TOTAL overprocessing saw an increase of 1.08 %, resulting in 5.98 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 3200115 (+1.60 %). These movements stand out and may require further investigation.

Average change by component:

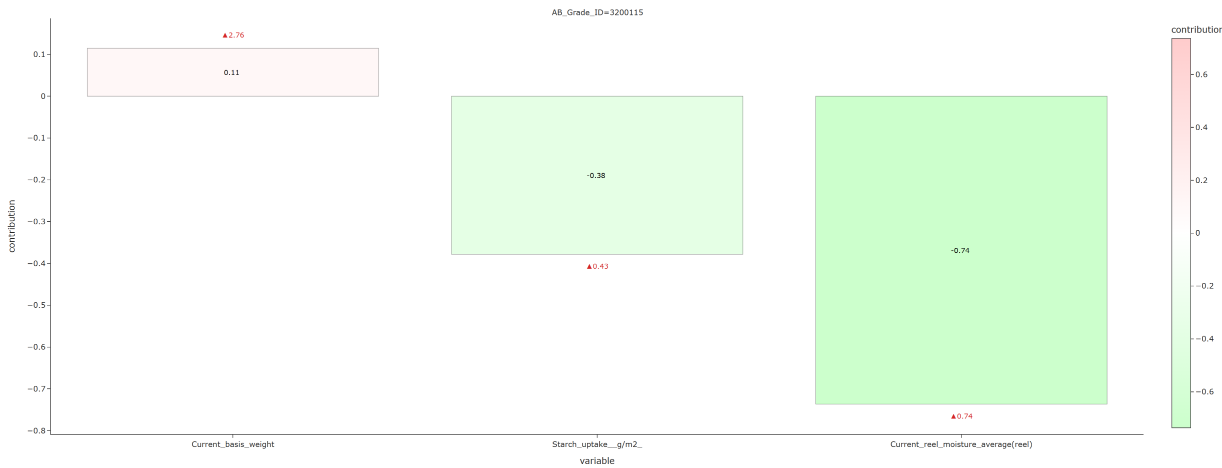
- **Burst** increased on average (+2.92 %).
- **CMT30** increased on average (+2.13 %).
- **SCT_CD** increased on average (+0.16 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+2.92 % on average), with the highest increase in grade 3200115 (+2.92 %). These components stand out and may require further investigation.

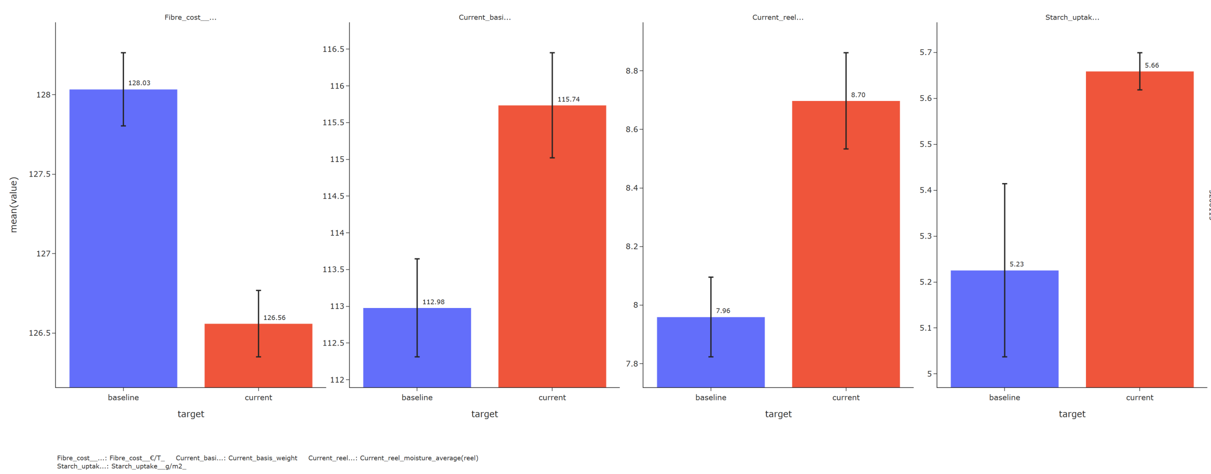
3 Breakdown by grades

3.1 3200115

3.1.1 Fiber cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Current reel moisture average(reel) increased (+0.74)



3.1.2 Steam cost

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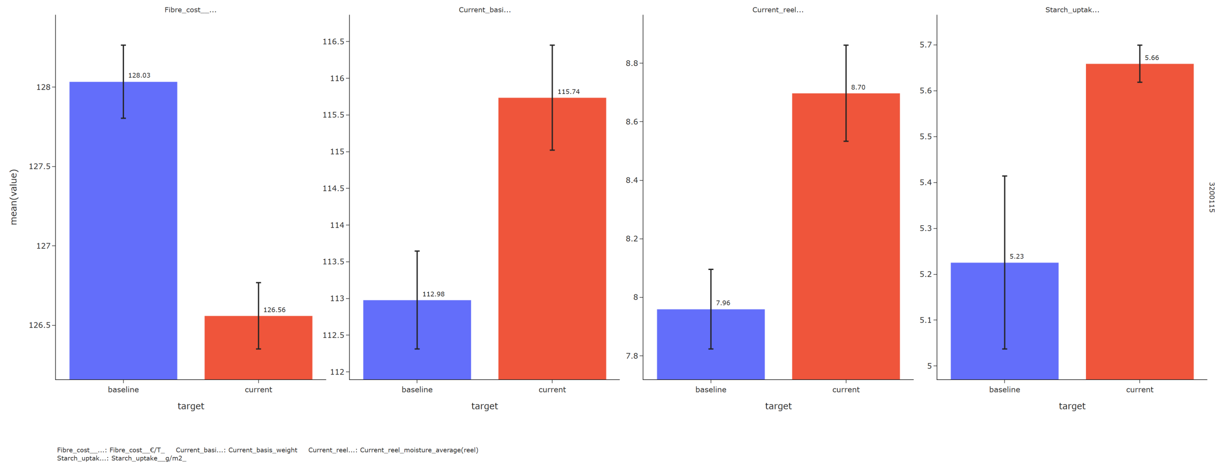
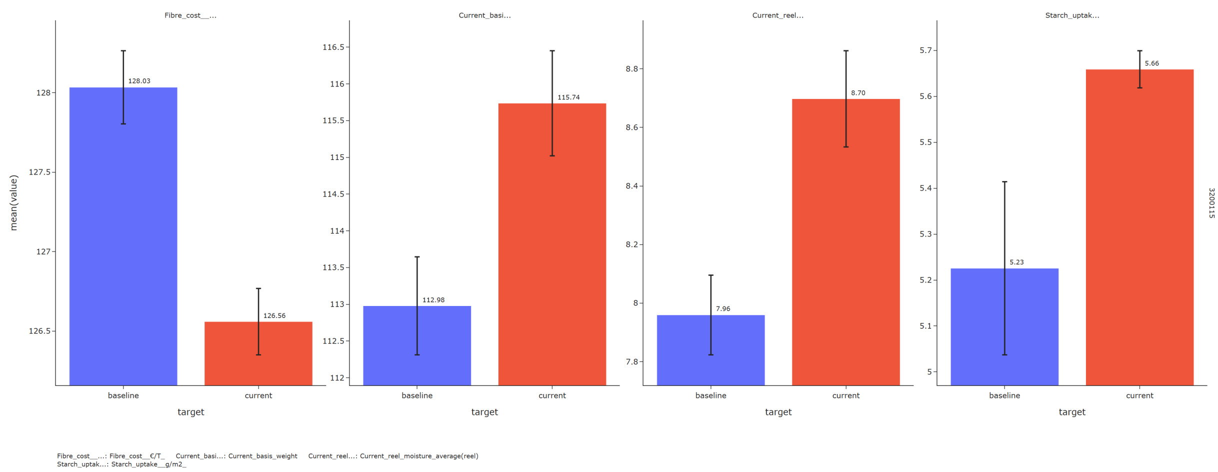
## KeyError: "['Steam__kWh/T__L1'] not in index"
```



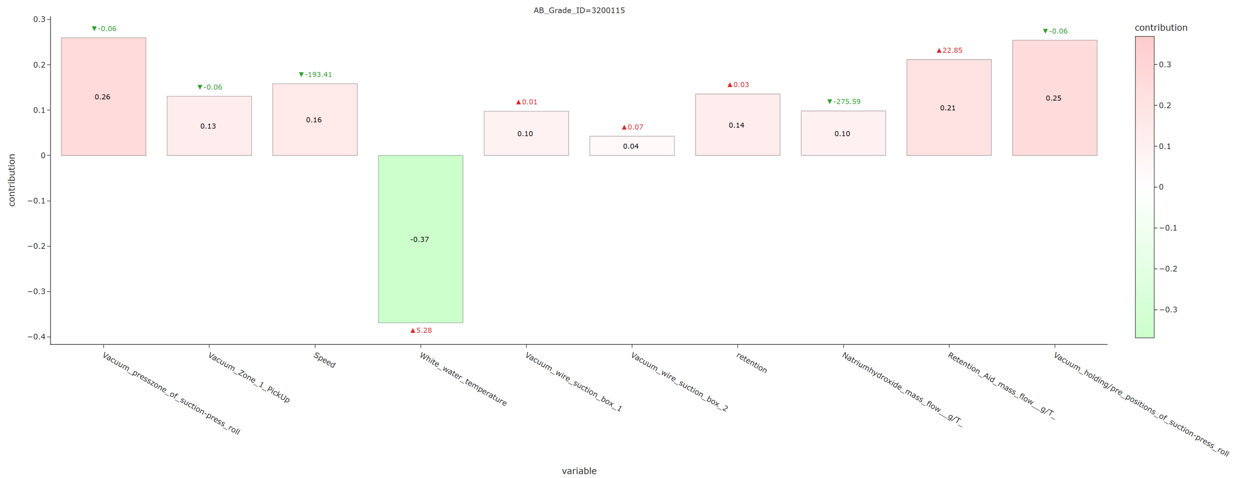
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Current reel moisture average(reel) increased (+0.74)

```
## KeyError: "['Steam__€/T_'] not in index"
```

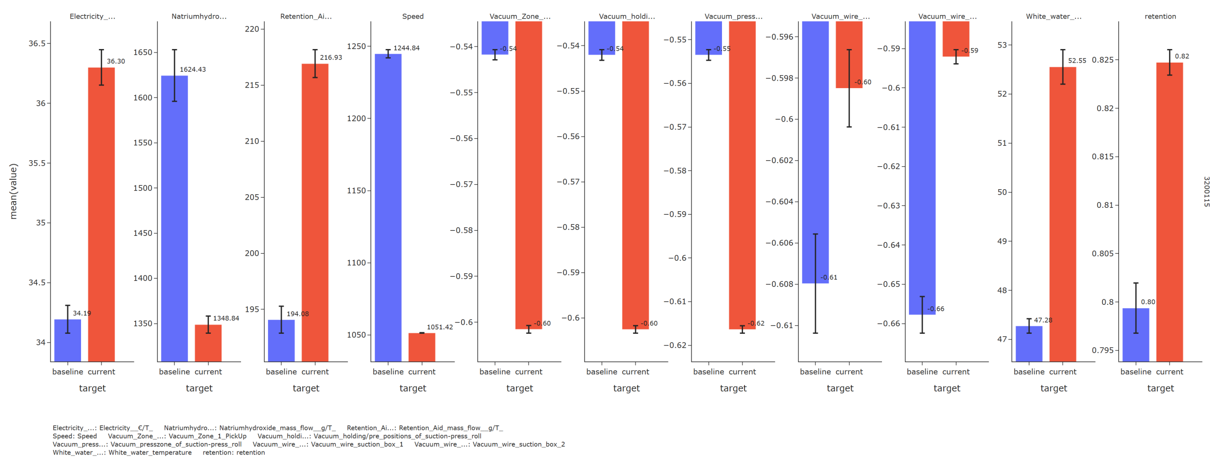


3.1.3 Electricity cost

```
## Conditional SHAP:   0%|          | 0/100 [00:00<?, ?it/s]Conditional SHAP:   2%|2         |
```



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - White water temperature increased (+5.28) **Cost-increasing drivers**: - Vacuum presszone of suction-press roll decreased (-0.06) - Vacuum holding/pre positions of suction-press roll decreased (-0.06)



3.1.4 Starch cost

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## Conditional SHAP:   0%|          | 0/100 [00:00<?, ?it/s]Conditional SHAP:   1%|1          |
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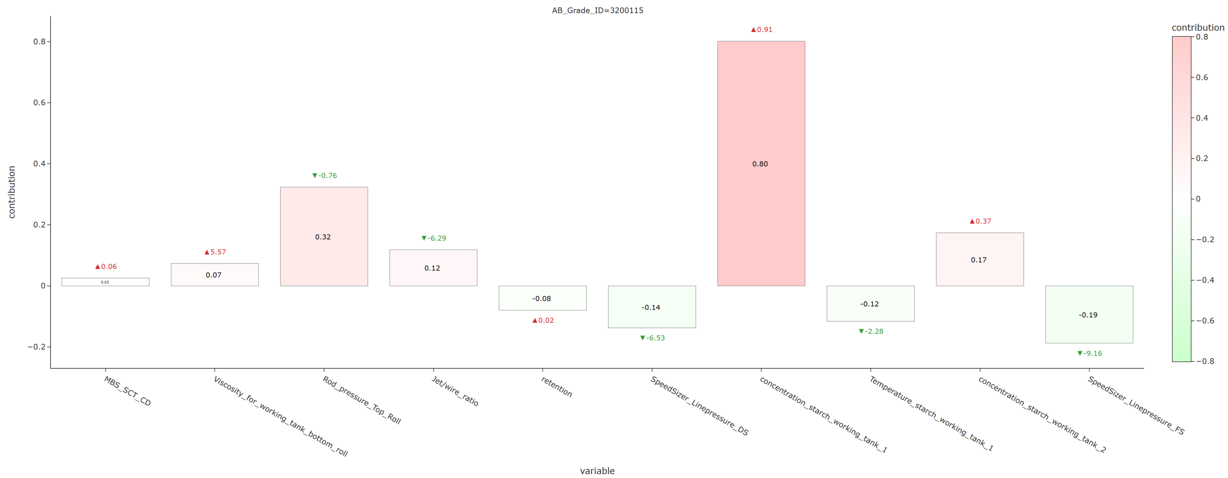
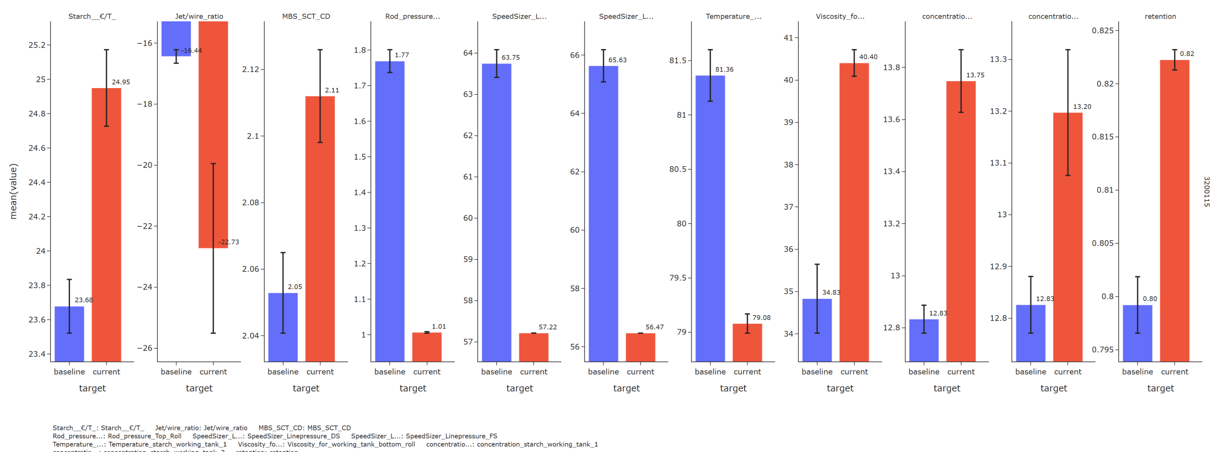
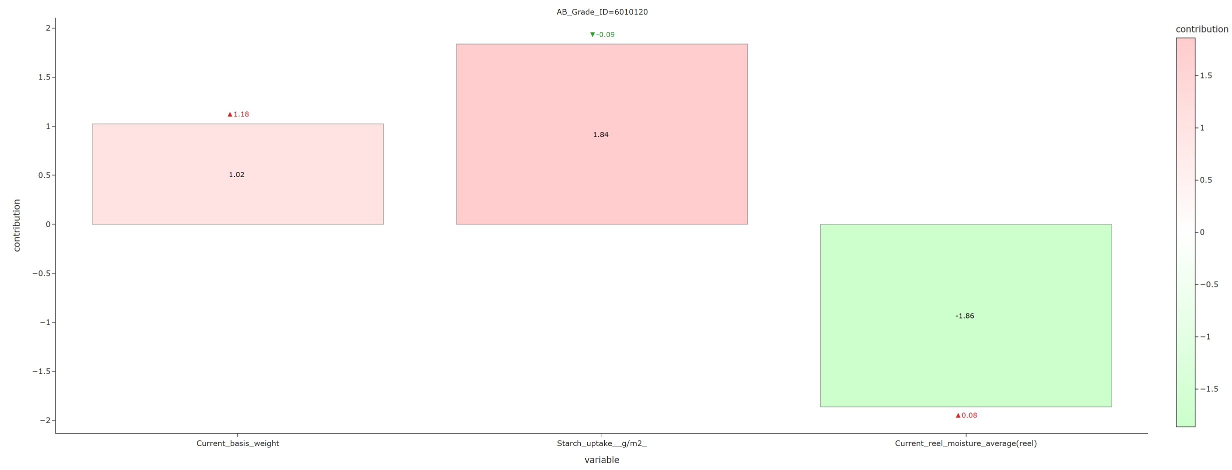


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - SpeedSizer Linepressure FS decreased (-9.16) **Cost-increasing drivers**: - Concentration starch working tank 1 increased (+0.91) - Rod pressure Top Roll decreased (-0.76)

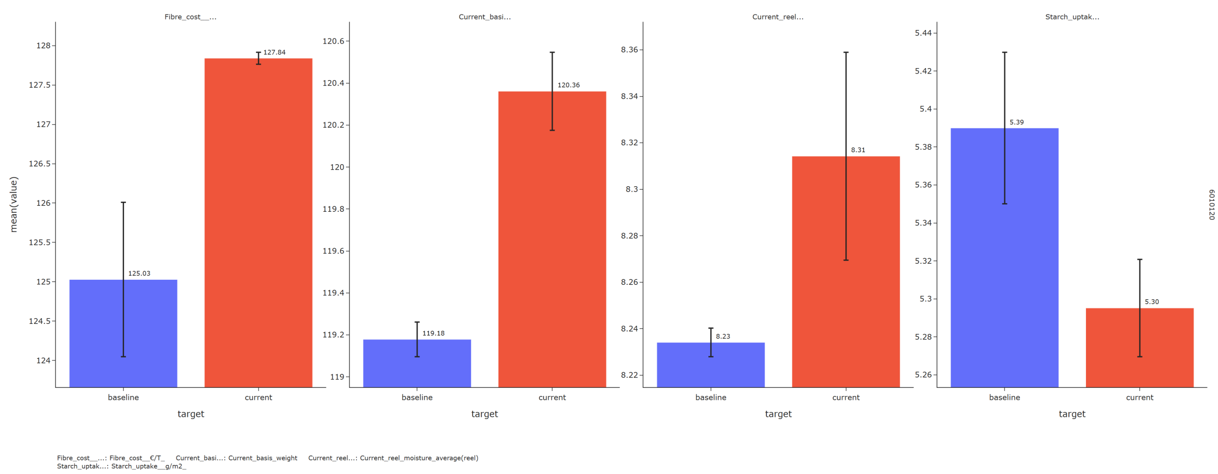


3.2 6010120

3.2.1 Fiber cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Current reel moisture average(reel) increased (+0.08)



3.2.2 Steam cost

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## KeyError: "['Steam__kWh/T__L1'] not in index"
```
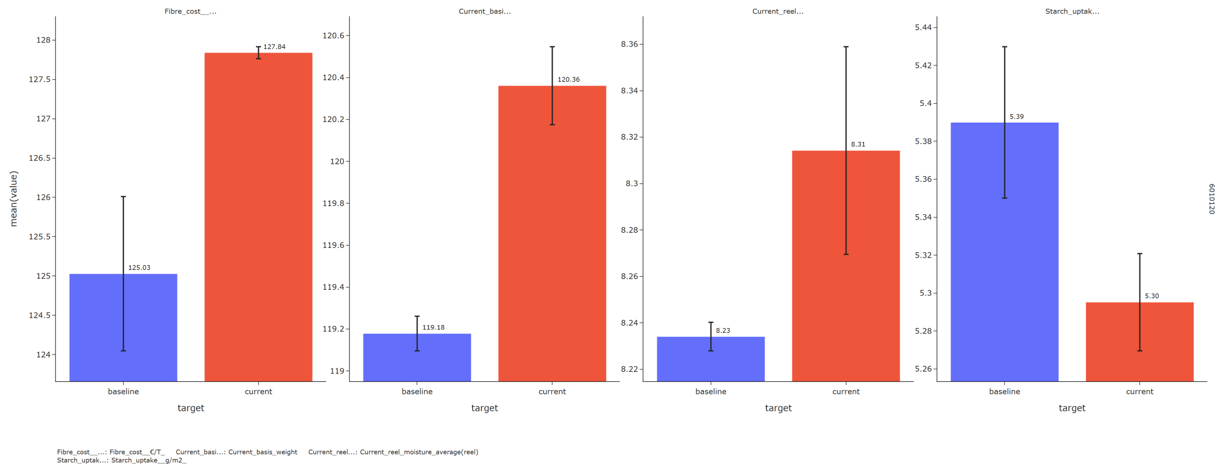
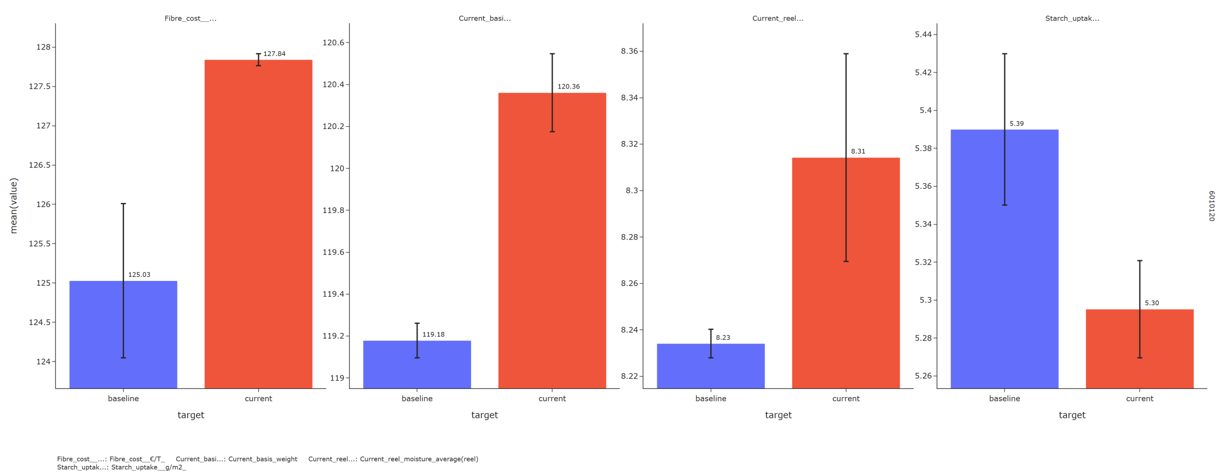



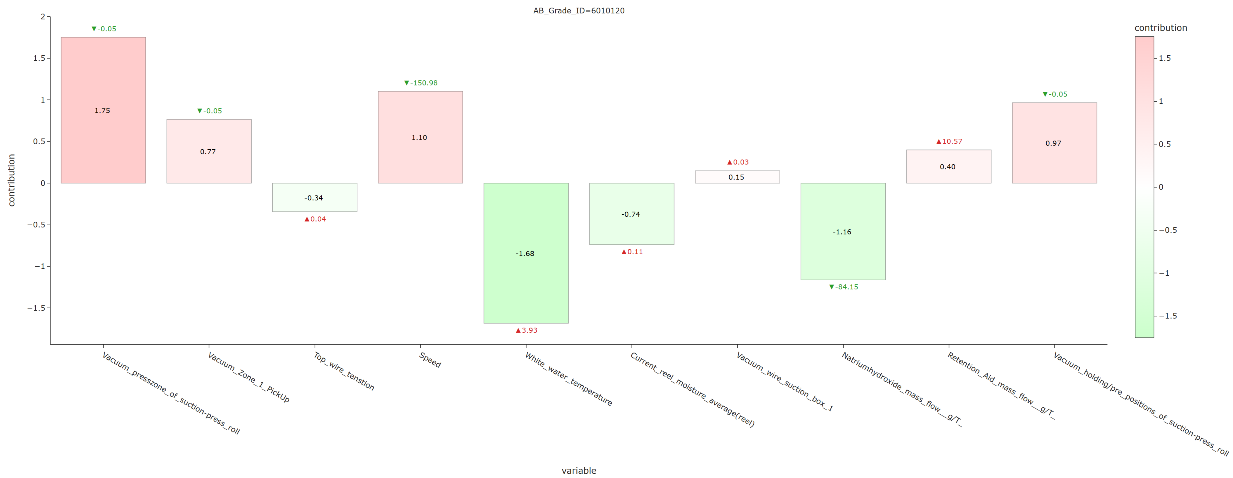
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - Current reel moisture average(reel) increased (+0.08)

```
## KeyError: "['Steam__€/T_'] not in index"
```

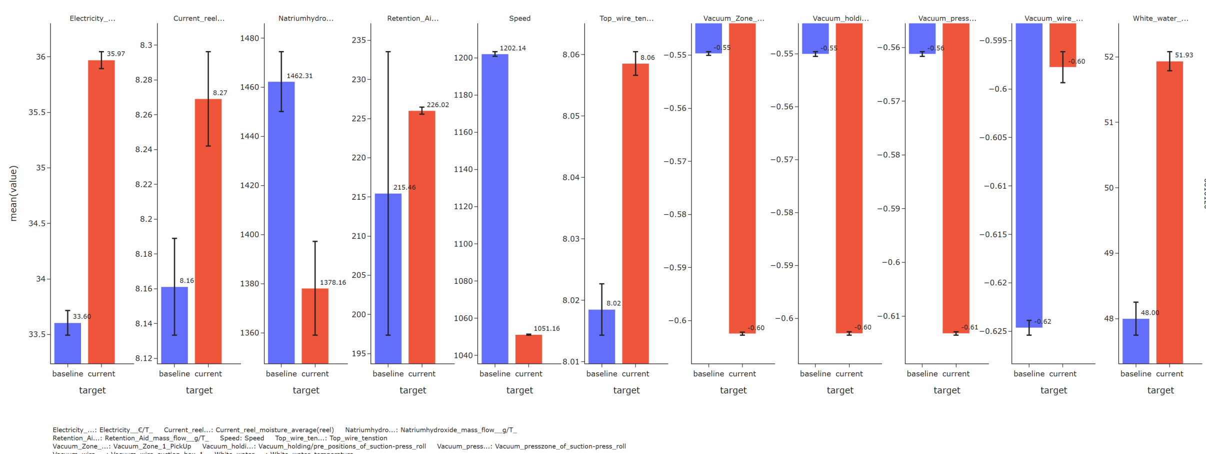


3.2.3 Electricity cost

```
## Conditional SHAP:   0%|          | 0/100 [00:00<?, ?it/s]Conditional SHAP:   1%|1         |
```

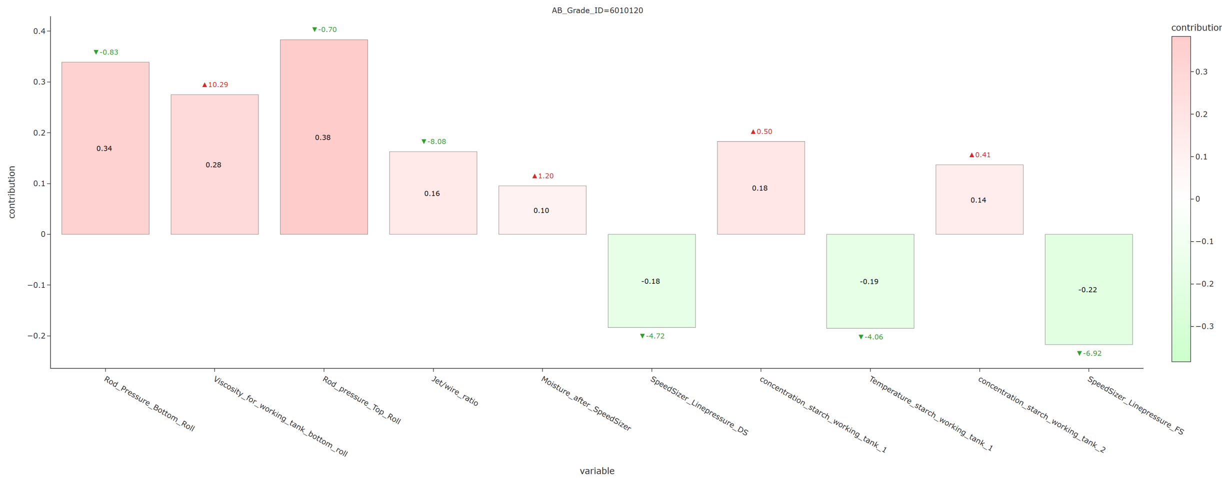


For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-reducing drivers**: - White water temperature increased (+3.93) - Natriumhydroxide mass flow g/T decreased (-84.15) **Cost-increasing drivers**: - Vacuum presszone of suction-press roll decreased (-0.05)



3.2.4 Starch cost

```
## Conditional SHAP:   0%|          | 0/100 [00:00<?, ?it/s]Conditional SHAP:   1%|1          |
```



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): **Cost-increasing drivers**: - Rod pressure Top Roll decreased (-0.70) - Rod Pressure Bottom Roll decreased (-0.83) - Viscosity for working tank bottom roll increased (+10.29)